

Strict inequalities for products of filters

OPG-37339

L1 - TRACTABLE

PARTIALLY SOLVED

Topology

ORIGINAL PROBLEM STATEMENT

Conjecture $\mathcal{A} \times_F^{\text{RLD}} \mathcal{B} \subset \mathcal{A} \ltimes \mathcal{B} \subset \mathcal{A} \times^{\text{RLD}} \mathcal{B}$ for some filter objects $\mathcal{A}, \mathcal{B}$. Particularly, is this formula true for $\mathcal{A} = \mathcal{B} = \Delta \cap \uparrow^{\mathbb{R}} (0; +\infty)$?

A weaker conjecture:

Conjecture $\mathcal{A} \times_F^{\text{RLD}} \mathcal{B} \subset \mathcal{A} \ltimes \mathcal{B}$ for some filter objects $\mathcal{A}, \mathcal{B}$.